

Supplementary Material

QSMEF: A Methodological Framework for Functional Analysis of Quantum Implementations

Additional Case Study Quantum Phase Estimation (QPE) in Shor’s Algorithm

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Abstract

This supplementary document complements the paper *QSMEF: A Methodological Framework for Functional Analysis of Quantum Implementations*. It presents the complete Quantum Phase Estimation (QPE) case study used to evaluate QSMEF in the context of Shor’s algorithm, including the implementation modeling, functional decomposition, observable definition, experimental results, and analysis of a modified implementation. This material is provided to support reproducibility and to make the extended evaluation available beyond the page-limited conference version of the paper.

1 Introduction

This supplementary document presents the second case study used to evaluate the QSMEF framework. Whereas the conference paper reports the evaluation based on the Shenvi–Kempe–Whaley (SKW) coined quantum walk search algorithm on the hypercube using the Grover coin and the flip-flop shift operator, this document presents the complete evaluation of the Quantum Phase Estimation (QPE) subroutine used in Shor’s algorithm.

The study follows the methodology presented in the main paper. QSMEF is applied to quantify the functional contribution of the implementation components through Hermitian observables and Shapley values. In addition, the case study analyzes the impact of a localized modification on the functional contribution of the different components.

2 Case Study: Quantum Phase Estimation (QPE) in Shor’s Algorithm

Modeling and Architecture. The system comprises two quantum registers: a counting register and a work register. The associated Hilbert space is defined as

$$\mathcal{H} = \mathcal{H}_{\text{count}} \otimes \mathcal{H}_{\text{work}}, \quad (1)$$

where $\mathcal{H}_{\text{count}}$ corresponds to the counting register, consisting of m qubits ($Q = 2^m$ computational basis states), and $\mathcal{H}_{\text{work}}$ corresponds to the work register. The system dynamics are governed by the modular unitary operator

$$U|y\rangle = |ay \bmod N\rangle, \quad (2)$$

where a is an integer coprime to N , and r is the period to be determined, defined by

$$a^r \equiv 1 \pmod{N}.$$

The eigenvectors of U satisfy

$$U|u_s\rangle = e^{2\pi i s/r} |u_s\rangle, \quad s \in \{0, \dots, r-1\}. \quad (3)$$

Each eigenvector $|u_s\rangle$ is associated with the phase s/r , which contains the information required to recover the period r through quantum phase estimation.

The work register is initialized in the state

$$|1\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle.$$

Consequently, the algorithm's success probability corresponds to the average over all possible phases s/r . The following section decomposes this implementation into the functional components analyzed by QSMEF.

Component Decomposition. For this case study, the functional components defined by QSMEF correspond to the main stages of the QPE implementation:

$$B_{\text{QPE}} = \{B_0, B_1, B_{2,0}, B_{2,1}, B_{2,2}, B_{2,3}, B_{2,4}\}. \quad (4)$$

Here, B_0 represents the initial state preparation ($|0\rangle^{\otimes m} \otimes |1\rangle$), B_1 groups the Hadamard gates applied to the counting register, and $B_{2,j}$ corresponds to the controlled modular exponentiation operation U^{2^j} for $j = 0, \dots, 4$.

The inverse Quantum Fourier Transform (QFT †) and the measurement stage belong to the evaluation procedure and are therefore not considered attributable functional components. The adopted partition follows a functional-stage granularity, separating the initial state preparation, the generation of superposition, and each controlled modular exponentiation operation.

The observable-based functional metric used by QSMEF is defined as

$$M_H(\rho) = \text{Tr}(H\rho), \quad (5)$$

where H is a Hermitian observable and ρ is the quantum state under evaluation.

For each coalition C of functional components, the characteristic function is defined as

$$v(C) = M_H(\rho_C) - M_H(\rho_\emptyset), \quad (6)$$

where $\rho_\emptyset$ denotes the state associated with the empty coalition.

QSMEF computes the functional contribution of each component using the Shapley value

$$\phi_b = \sum_{C \subseteq B \setminus \{b\}} \frac{|C|!(|B| - |C| - 1)!}{|B|!} [v(C \cup \{b\}) - v(C)], \quad (7)$$

which satisfies the efficiency property

$$\sum_{b \in B} \phi_b = v(B). \quad (8)$$

Functional Observable. The functional evaluation is performed using the projective observable H_{per} , defined over the counting register. Let $\mathcal{K}_r$ denote the set of outcomes compatible with the period r within a tolerance ε . Then,

$$H_{\text{per}} = \left(\sum_{k \in \mathcal{K}_r} |k\rangle\langle k| \right) \otimes I_{\text{work}}, \quad (9)$$

where I_{work} denotes the identity operator acting on the work register.

This observable represents the algorithm's operational success criterion, as it projects onto the subspace associated with outcomes compatible with the target period.

Applying the functional metric defined in Eq. (5) to the observable H_{per} yields a measure that coincides with the probability of observing an outcome belonging to $\mathcal{K}_r$. The extension by the identity operator reflects that the evaluation criterion depends exclusively on the counting register.

Implementation Results. Figure 1 presents the Shapley values obtained for the QPE implementation with $N = 21$, $a = 2$, and $m = 5$. The phase estimation tolerance was set to $\varepsilon = 1/2^5$, according to the number of qubits in the counting register.

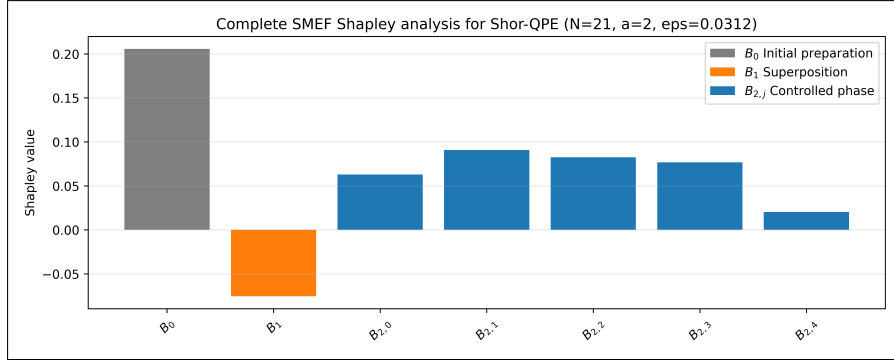


Figure 1: Shapley values for the QPE implementation with $N = 21$ and $a = 2$.

The negative value associated with B_1 does not imply that the Hadamard gates are unnecessary. Since the Shapley value measures the average marginal contribution over all possible coalitions, this result indicates that B_1 , when considered in isolation, does not increase the success probability captured by H_{per} . Nevertheless, the Hadamard gates are essential for generating the superposition on which the controlled modular exponentiation operations act. Their contribution arises primarily through their interaction with these components, revealing the cooperative nature of the implementation.

Table 1 reports the Shapley values for each functional component together with its relative contribution to the total game value.

The sum of the Shapley values equals the total game value, thereby confirming the efficiency property (Eq. 8). Using $M_{H_{\text{per}}}(\rho_B) \approx 0.745$ and $M_{H_{\text{per}}}(\rho_\emptyset) \approx 0.281$, we obtain $v(B) \approx 0.463487$, which exactly matches the sum of the contributions reported in Table 1.

Table 1: Shapley values and relative contribution of each functional component in QPE.

Component	ϕ_b	$\phi_b/v(B)$ (%)
B_0	0.205753	44.39
B_1	-0.075497	-16.29
$B_{2,0}$	0.062984	13.59
$B_{2,1}$	0.090666	19.56
$B_{2,2}$	0.082468	17.79
$B_{2,3}$	0.076805	16.57
$B_{2,4}$	0.020308	4.38
Total	0.463487	100.00

Evaluation of a Modified Implementation. Figure 2 compares the Shapley values obtained for the original implementation and a modified implementation. The modification consists of replacing, within component $B_{2,2}$, the unitary operator constructed with parameter $a = 2$ by the corresponding operator with $a = 8$.

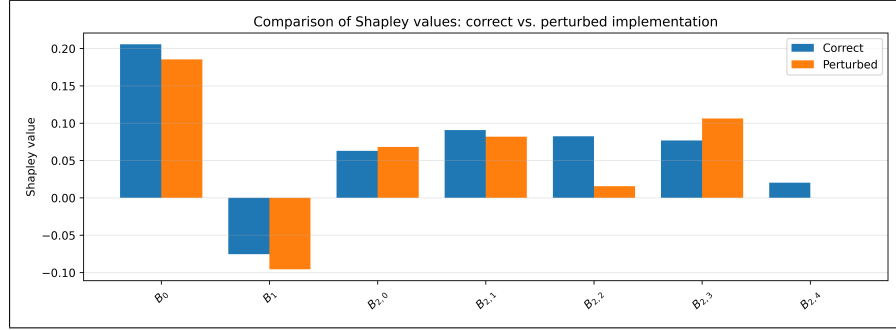


Figure 2: Comparison of the Shapley values for the original and modified implementations.

Table 2 reports the Shapley values (Eq. 7) obtained for the original and modified implementations, together with the difference between both scenarios. The *Total* row confirms the efficiency property (Eq. 8).

Table 2: Comparison of the Shapley values before and after modifying component $B_{2,2}$.

Component	Original	Modified	Difference
B_0	0.205753	0.185445	-0.020308
B_1	-0.075497	-0.095805	-0.020308
$B_{2,0}$	0.062984	0.068229	0.005246
$B_{2,1}$	0.090666	0.081870	-0.008796
$B_{2,2}$	0.082468	0.015579	-0.066889
$B_{2,3}$	0.076805	0.106278	0.029473
$B_{2,4}$	0.020308	0.000000	-0.020308
Total	0.463487	0.361596	-0.101891

Modifying component $B_{2,2}$ reduces its Shapley value from 0.082 to 0.016, corresponding to an approximate decrease of 81%. Moreover, the modification alters the functional contribution of other components: the contribution of $B_{2,3}$ increases, whereas the Shapley value of $B_{2,4}$ decreases to zero.

The sum of the Shapley values decreases from 0.463487 to 0.361596, reflecting a reduction in the

total game value $v(B)$, in agreement with the efficiency property defined in Eq. 8. This variation indicates that modifying the unitary operator reduces the overall functional contribution of the implementation with respect to the observable H_{per} .

These results show that modifying a single component changes the functional contribution of other components in the circuit. The proposed framework enables the comparison of different implementations and quantifies how such modifications affect the functional contribution of each component.

3 Conclusions

This supplementary case study extends the experimental evaluation of QSMEF through the Quantum Phase Estimation (QPE) subroutine used in Shor’s algorithm. The results show that the framework quantifies the functional contribution of individual implementation components while capturing the cooperative effects arising from their interaction.

The analysis of the modified implementation demonstrates that a localized change in one controlled modular exponentiation component affects not only its own functional contribution but also the contributions of other components in the circuit.

Together with the evaluation of the Shenvi–Kempe–Whaley (SKW) coined quantum walk search algorithm on the hypercube using the Grover coin and the flip-flop shift operator presented in the conference paper, this case study provides additional evidence of the applicability of QSMEF to quantum implementations with different computational structures.